

A MULTIDISCIPLINARY APPROACH TO TOXICOLOGICAL SCREENING: III. NEUROBEHAVIORAL TOXICITY

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The neurobehavioral effects of 10 known toxicants were examined as part of a multidisciplinary screening battery. The toxicants included carbaryl (CAR), triadimefon (TDM), heptachlor (HEP), chlordane (CDN), diethylhexyl phthalate (DEHP), carbon tetrachloride (CCl₄), phenol, trichloroethylene (TCE), tetrachloroethylene (PER or perchlorethylene), and dichloromethane (DCM or methylene chloride). A functional observational battery and motor activity measurements were conducted before exposure, at specified times after an acute exposure, and during and after 14-d exposure. Severity scoring analysis was used to generate profiles of effect. The pesticides, CAR, TDM, HEP, and CDN, displayed the most acute neurotoxicity and were active at lower proportions of their respective acute LD50 values than were the solvents or the industrial chemicals. Although CAR and TDM showed little or no neurobehavioral effects with repeated dosing, cumulative neurotoxicity and lethality were evident with HEP and CDN. Phenol produced acute convulsive effects, and the most prominent finding with repeated exposure was lethality. DEHP displayed no neurobehavioral toxicity. The organic solvents, TCE, PER, CCl₄, and DCM, produced various degrees of general nervous system depression following acute administration of high dose levels. Repeated dosing produced little or no effect with TCE or PER, marked physiological changes with CCl₄, and cumulative toxicity and lethality with DCM. Some results of these studies were unexpected and should provide impetus for further research. Overall, these findings illustrate the utility of these screening methods.

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Despite an increasingly large number of chemicals in the environment, the potential noncancer toxicity of the majority of these chemicals is not known (National Research Council, 1984). Efforts have been underway to develop a strategy for the evaluation and generation of a database for these chemicals. Our series of reports (Berman et al., 1995; MacPhail et al., 1995; Narotsky & Kavlock, 1995) describes an integrated protocol designed to detect and quantify chemical-induced neurotoxicity, systemic toxicity, and developmental toxicity. This article describes the neurotoxic evaluation of 10 chemicals following a single oral dose (acute) and during short-term repeated oral dosing (14 d). Nine of these chemicals are found at sites on the Superfund National Priorities List (U.S. EPA, 1987) and may pose a significant threat to human health. The chemicals included four pesticides (carbaryl, triadimefon, heptachlor, and chlordane), four chlorinated solvents (tetrachloroethylene, trichloroethylene, carbon tetrachloride, and dichloromethane), and two industrial chemicals (phenol and diethylhexyl phthalate).

A neurobehavioral screening battery consisting of a functional observational battery (FOB) and automated motor activity measurements (MacPhail et al., 1989; Moser et al., 1988; Moser, 1989) was used to evaluate the neurotoxic potential of these chemicals. The FOB consists of a systematic assessment of home-cage and open-field behaviors and neurological function of rats. Automated motor activity assessment provides an objective apical measure of neurobehavioral function. These screening methods are consistent with the recommendations of several expert panels for first-tier testing of chemicals (reviewed in Office of Technology Assessment, 1990). Such tests are required by the U.S. Environmental Protection Agency (EPA) for testing of pesticides under the Federal Insecticide, Fungicide, and Rodenticide Act (FIFRA) and for test rules and consent agreements developed under the Toxic Substances Control Act (TSCA) (U.S. EPA, 1991). This battery of tests is primarily intended to provide data on dose-response and time-course characteristics of a chemical, and neurological profiles of effect that can then be used to make decisions concerning neurotoxic potential and lead to subsequent studies for risk assessment and risk management (Moser, 1991b).

In the present study, neurobehavioral evaluations and systemic toxicity assessments were conducted in the same rats. Immediately after the last neurobehavioral tests, rats were killed and tissues excised for evaluation of organ weights, serum chemistry, urine specific gravity, and histopathological alterations. Developmental toxicity was evaluated in separate groups of rats exposed during gestation (d 6–20), using the method described by Chernoff and Kavlock (1982), which examined pre- and postnatal growth and viability in the offspring, as well as maternal effects. Systemic and developmental toxicity data are presented elsewhere (Berman et al., 1995; Narotsky & Kavlock, 1995).

METHODS

Chemicals

Carbaryl (CAR; 99%) was kindly donated by Rhone-Poulenc AG Co. (Research Triangle Park, N.C.), and triadimefon (TDM; Technical Bayleton;

93.4%) was donated by Mobay Chemical Co. (Kansas City, Mo.). The technical products of heptachlor (HEP; approximately 73% heptachlor/26% chlordane) and chlordane (CDN; approximately 40% chlordane/60% other cyclodiene products) were donated by Velsicol Chemical Corp. (Rosemont, Ill., and Memphis, Tenn.). Analytical grade (>99%) diethylhexyl phthalate (DEHP), carbon tetrachloride (CCl₄), phenol, trichloroethylene (TCE), tetrachloroethylene (PER; perchlorethylene), and dichloromethane (DCM; methylene chloride) were obtained from Aldrich Chemical Co. (Milwaukee, Wis.). Phenol was mixed with deionized water, and triadimefon was suspended in 7.5% Emulphor EL-620 (GAF Corp., Charlotte, N.C.) and water. All other chemicals were dissolved in corn oil (Mazola).

Dose selection was based on the estimated acute LD50 values (see Berman et al., 1995). The upper limit of dose was 5 g/kg. In the acute study, treatment was either vehicle, or 3%, 10%, 30%, or 56% of the LD50, or 3%, 10%, 30%, or 100% of the limit dose (if the LD50 could not be determined). In the subacute study the dose groups were vehicle, and 1%, 3%, 10%, or 30% of the LD50 value (or limit dose). However, for triadimefon the dose specification was 1%, 3%, 10%, or 30% in the acute study, and 0.1%, 0.3%, 3%, or 10% in the subacute study. Table 1 lists the dose levels and volumes used for each chemical. In all experiments, *n* = 8 rats/dose level.

Animals

Adult female Fischer 344 rats (CRL-CD-BR; Charles River Laboratories, Raleigh, N.C.) were used in all studies. Rats were individually housed in polycarbonate cages (26.5 × 24 × 20 cm) with hardwood chip bedding. Deionized tap water and standard rat chow (Purina rodent chow 5002) were

TABLE 1. Chemical Treatments for Each Experiment, Including Dose Levels, Vehicle, and Volume of Administration

Chemical	Acute exposure	Repeated exposure	Vehicle (dose volume)
Pesticides			
CAR	8, 26, 78, 145 mg/kg	3, 8, 26, 78 mg/kg	Corn oil (1 ml/kg)
TDM	3, 10, 32, 96 mg/kg	0.3, 1, 10, 32 mg/kg	Emulphor/water (1 ml/kg)
HEP	7, 23, 69, 129 mg/kg	2, 7, 23, 69 mg/kg	Corn oil (1 ml/kg)
CDN	16, 52, 156, 291 mg/kg	5, 16, 52, 156 mg/kg	Corn oil (1 ml/kg)
Industrial chemicals			
Phenol	12, 40, 120, 224 mg/kg	4, 12, 40, 120 mg/kg	Deionized water (1 ml/kg)
DEHP	150, 500, 1500, 5000 mg/kg	50, 150, 500, 1500 mg/kg	Corn oil (10 ml/kg)
Solvents			
PER	150, 500, 1500, 5000 mg/kg	50, 150, 500, 1500 mg/kg	Corn oil (10 ml/kg)
TCE	150, 500, 1500, 5000 mg/kg	50, 150, 500, 1500 mg/kg	Corn oil (10 ml/kg)
CCl ₄	150, 500, 1500, 5000 mg/kg	50, 150, 500, 1500 mg/kg	Corn oil (10 ml/kg)
DCM	101, 337, 1012, 1889 mg/kg	34, 101, 337, 1012 mg/kg	Corn oil (10 ml/kg)

Note. All treatments were administered orally. Acute LD50 values were determined for each chemical initially. Dose levels were set as 3% to 56% of the LD50 value, or 3% to 100% of the limit dose, for acute studies, and 1% to 30% of the LD50 or limit dose for subacute studies (see Methods).

available ad libitum. The animal room temperature was maintained at $21 \pm 1^\circ\text{C}$, with $55 \pm 5\%$ humidity, and had a 12-h light/dark cycle (lights on at 0630). Animals were allowed at least 1 wk after arrival for quarantine and acclimation to the facility before testing began. The rats were identified by ear tag and assigned to test groups using random stratification tables based on body weights.

Behavioral Evaluations

Functional Observational Battery Individual measures comprising the FOB are listed in Table 2. Technical procedures and scoring criteria are described in detail elsewhere (McDaniel & Moser, 1993). Home-cage observations included descriptions of posture, palpebral closure, and convulsions or tremors in the rat. The rat's reactivity to being removed and handled, and any salivation, lacrimation, piloerection, and ptosis, were ranked. The rat was then placed in an open field (the top of a 60×90 cm laboratory cart with a 6.5-cm rim) for 3 min, during which time arousal level (alertness) and gait abnormalities were scored, the presence of any convulsions or tremors was noted, and the numbers of rears, fecal boluses, and urine pools were counted. The reactions to several stimuli were scored: the approach of a pencil, touch on the rump with a blunt instrument, the sound of a metal clicker, and tail pinch using forceps. Pupillary response was tested using a penlight stimulus. Aerial righting reflex was rated. Forelimb and hindlimb grip strengths were measured using strain gauges, after which the hind feet were painted lightly and the rat was dropped onto paper to measure landing foot splay. Finally, the rat was weighed and rectal body temperature was taken. All data were recorded on standardized sheets and later entered into a computer for further analysis.

Motor Activity Motor activity was measured in a maze composed of a series of interconnected alleys shaped like a figure eight with two blind

TABLE 2. Measures of the Neurobehavioral Test Battery, Organized According to Domains of Neurological Function

Autonomic	Activity	Excitability
Lacrimation	Rearing	Ease of removal
Salivation	Motor activity counts	Handling reactivity
Palpebral closure	Home cage posture	Arousal
Pupil response		Clonic movements
Urination		Tonic movements
Defecation		
Neuromuscular	Sensorimotor	Physiological measures
Gait score	Tail-pinch response	Body weight
Righting reflex	Click response	Body temperature
Forelimb grip strength	Touch response	Piloerection
Hindlimb grip strength	Approach response	
Landing foot splay		

alleys projecting from opposite sides of a central arena (Reiter, 1983). Six pairs of phototransmitter/diode detectors were equally spaced around the figure eight portion of the maze, and one pair was located in each blind alley. Motor activity was recorded by a microprocessor as photocell interruptions during 1-h sessions.

Experimental Design The FOB only was performed on each rat just prior to the first dose (to provide baseline values). Thereafter, the FOB followed by motor activity assessments was conducted in each rat. For the acute study, testing took place at approximately 4 and 24 h after exposure. In the 14-d repeated exposure study, testing was conducted on d 4 and 9 (before the daily dose) and approximately 24 h after the 14th dose (d 15). All testing was performed "blind" such that the observer did not know the dose level.

Data Analysis A severity scoring scheme was used to determine chemical effects on specific domains of neurological function (Table 2). This method of analysis normalized individual data for all measures to a 1-to-4 scale (McDaniel & Moser, 1993; Moser, 1991b). The conversion of individual data to a severity score was based on the data for control rats at each time point. In general, a score of 1 reflected what was observed in most control rats, whereas a score of 4 reflected a rare occurrence in control rats. For continuous measures, score assignments were based on the range defined by the mean and standard deviation of the control group. Measures that were originally scored in terms of severity of effect (such as lacrimation, gait score, etc.) were converted directly to the severity scale. For other rank and quantal data, the score assignments were based on the frequency of occurrence of each score in the control rats and on the scalar distance from the mode. For each rat at each time point, these scores were then summed across the measures that comprised each functional domain. These composite scores were then averaged across treatment groups and examined across time, and were analyzed using an analysis of variance (ANOVA). For each functional domain that showed significant overall effects on either the dose factor or dose-by-time interaction, the individual measures that comprise that domain were then analyzed.

Subsequent analysis of individual FOB measures within a functional domain was carried out as described in Moser et al. (1988) and Creason (1989). In addition, physiological measures such as body weight and body temperature were analyzed for each study. Two-way ANOVAs were carried out using a grouping factor of dose, and repeated measures across time. Continuous data were analyzed by a linear model (GLM; SAS Institute, Inc., 1985) using each rat's baseline value as a covariate. Descriptive and rank data were analyzed using a categorical procedure (CATMOD; SAS Institute, Inc., 1985). Univariate analyses at each time point included group comparisons to control group values (SAS Institute, Inc., 1985). Resulting p values $\leq .05$ were considered significant.

RESULTS

Table 3 presents the results of the severity scoring analysis and indicates functional domains that were significantly affected for each chemical (i.e., significant overall dose or dose-by-time interaction in the ANOVA); these domains indicate the areas of concern for the toxicity of each chemical. Dose-response and time-course data for these domains are illustrated in Figures 1 through 9, as well as exemplary data for individual measures. Table 4 presents the no-effect levels (NOEL) determined for each chemical. Details of the effects of the various chemicals are described next.

TABLE 3. Results of Severity Scoring Analysis

Chemical	Functional domain				
	Autonomic	Activity	Excitability	Neuromuscular	Sensorimotor
Pesticides					
CAR, acute	+	+	—	—	+
CAR, subacute	—	—	—	—	—
TDM, acute	+	+	+	—	—
TDM, subacute	—	—	—	+	—
HEP, acute	—	+	+	—	—
HEP, subacute	+	+	+	—	—
CDN, acute	—	—	+	+	—
CDN, subacute	+	+	+	—	+
Industrial chemicals					
Phenol, acute	—	+	—	+	+
Phenol, subacute	+	+	—	—	—
DEHP, acute	—	—	—	—	—
DEHP, subacute	—	—	—	—	—
Solvents					
PER, acute	+	+	+	+	+
PER, subacute	—	—	—	+	—
TCE, acute	—	+	—	+	+
TCE, subacute	—	+	—	+	—
CCl ₄ , acute	—	+	—	+	—
CCl ₄ , subacute	—	+	—	—	—
DCM, acute	—	—	—	+	+
DCM, subacute	+	—	+	+	+

Note. Functional domains that showed a significant (as determined by ANOVA) increase in severity score are indicated by plus sign.

TABLE 4. No Observable Effect Level (NOEL) or Lowest Observable Effect Level (LOEL) for Each Chemical in the Acute and Subacute Studies

Chemical	NOEL	LOEL	Proportion of LD50
Pesticides			
CAR, acute	8 mg/kg		3%
CAR, 14-d	78 mg/kg ^a		30%
TDM, acute	10 mg/kg		3%
TDM, 14-d	32 mg/kg ^a		10%
HEP, acute		7 mg/kg ^b	3%
HEP, 14-d	2 mg/kg		1%
CDN, acute	52 mg/kg		10%
CDN, 14-d	5 mg/kg		1%
Industrial chemicals			
Phenol, acute	40 mg/kg		10%
Phenol, 14-d	12 mg/kg		3%
DEHP, acute	5000 mg/kg ^a		100% limit dose ^c
DEHP, 14-d	1500 mg/kg ^a		30% limit dose
Solvents			
PER, acute		150 mg/kg ^b	3% limit dose
PER, 14-d	1500 mg/kg ^a		30% limit dose
TCE, acute	500 mg/kg		10% limit dose
TCE, 14-d	150 mg/kg		3% limit dose
CCl ₄ , acute	150 mg/kg		3% limit dose
CCl ₄ , 14-d	150 mg/kg		3% limit dose
DCM, acute	337 mg/kg		10%
DCM, 14-d	101 mg/kg		3%

Note. The NOEL was determined to be the highest dose tested that did not produce significant behavioral changes. If no NOEL could be determined, the LOEL (lowest dose that produced significant changes) is listed.

^aHighest dose used.

^bLowest dose used.

^cLimit dose 5000 mg/kg.

Carbaryl

No treatment-related deaths occurred with CAR. Most of the effects of acute CAR were produced by the two highest doses (78 and 145 mg/kg) on the day of exposure; there were almost no effects obtained in the 14-d study.

Severity scoring analysis of the acute data revealed significant effects at 4 h in the autonomic, activity, and sensorimotor domains (Figure 1). Prominent autonomic signs included miosis, inhibited pupil response, and lacrimation. Activity changes consisted of decreased motor activity, and rats in the highest dose group displayed flattened posture in the home cage. Sensorimotor changes consisted of decreased responses to stimuli, most notably the tail

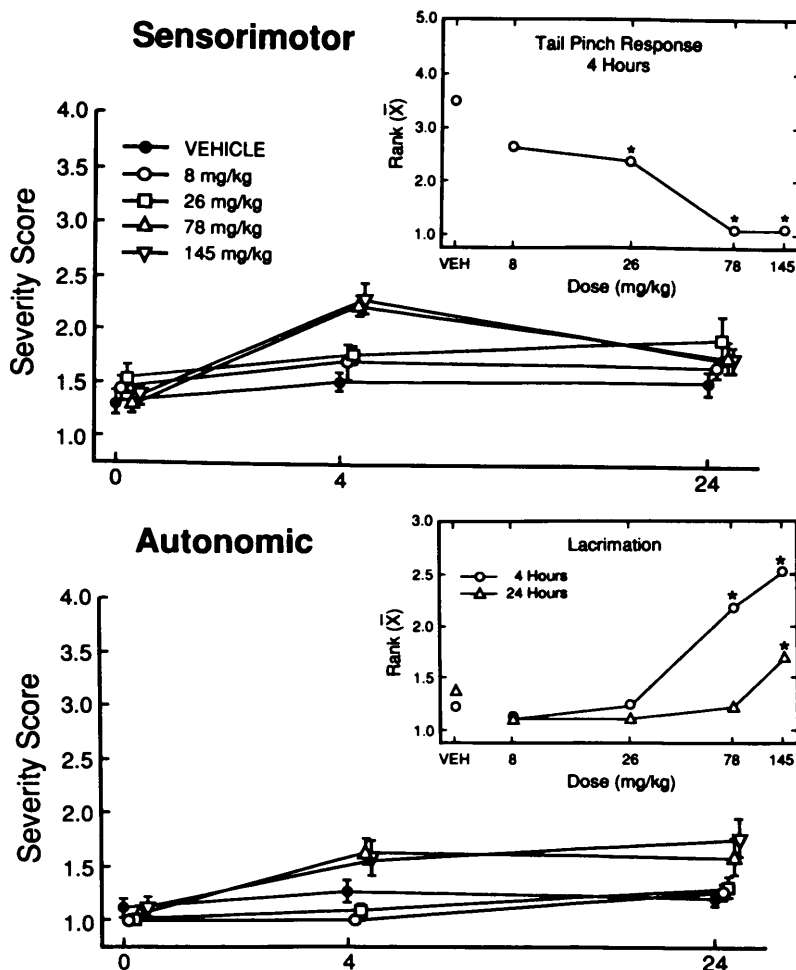


FIGURE 1. Dose response and time course of severity scores in each functional domain significantly affected by CAR (acute study). Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

pinch; in fact, rats in the two highest dose groups showed no visible response to the tail-pinch stimulus. Significant physiological alterations included weight loss (about 6%) and hypothermia (1.4°C decrease). Decreased tail-pinch response and weight loss were the most sensitive effects, significant in all but the lowest dose group. Clonic motor movements, consisting of smacking the mouth and mild whole-body tremors, were observed in most rats treated with the two highest doses.

The autonomic domain was also significantly affected at 24 h; residual effects included miosis, inhibited pupil response, and slight lacrimation.

No neurological domain was significantly affected with repeated dosing.

The only consistent finding was depressed body weight gain in the group receiving the highest dose (about 5% less than controls at the end of the study).

Triadimefon

No lethality occurred in either TDM study. All of the acute effects of TDM were produced by the two highest doses (32 and 96 mg/kg). No clear changes were observed in the 14-d study.

TDM produced significant changes in the autonomic, activity, and excitability domains at 4 h after dosing (Figure 2). Changes in excitability included increased arousal, tenseness, and increased reactivity to removal from the home cage. Stereotypic grooming and straub tail were also noted. Hyperactivity was prominent, and included increased activity in the figure-eight maze (up to 200% of vehicle control levels), and marked increases in rearing in the home cage and open field. Altered autonomic function included inhibited pupil response (mydriasis) and increased urination. Hyperthermia was also evident at 4 h (increase of about 1°C).

Significant effects at 24 h were limited to the highest dose group (96 mg/kg) and consisted of changes in excitability and activity. These effects included increased arousal, mild tremors and quivers, and increased rearing.

In contrast to the acute study, no functional domain was clearly affected by TDM with repeated dosing. Although effects on the neuromuscular domain were significant at 4 d in the two highest dose groups, no individual measures were affected. Closer examination of the severity scores indicated

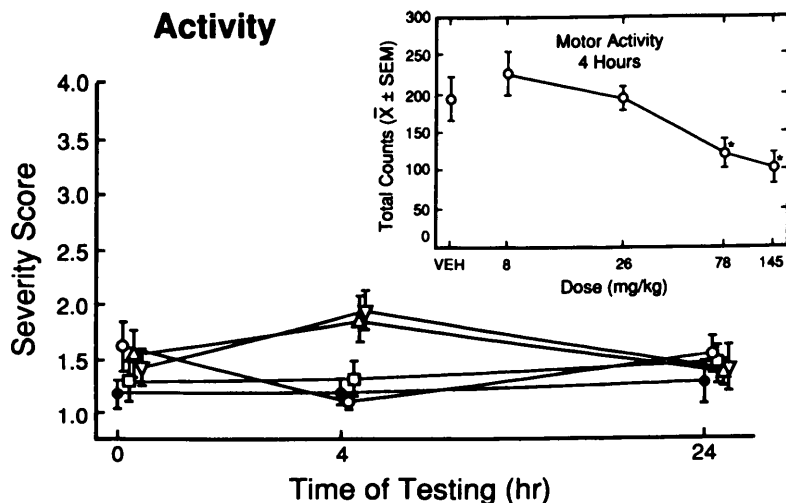


FIGURE 1. (Continued) Dose response and time course of severity scores in each functional domain significantly affected by CAR (acute study). Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

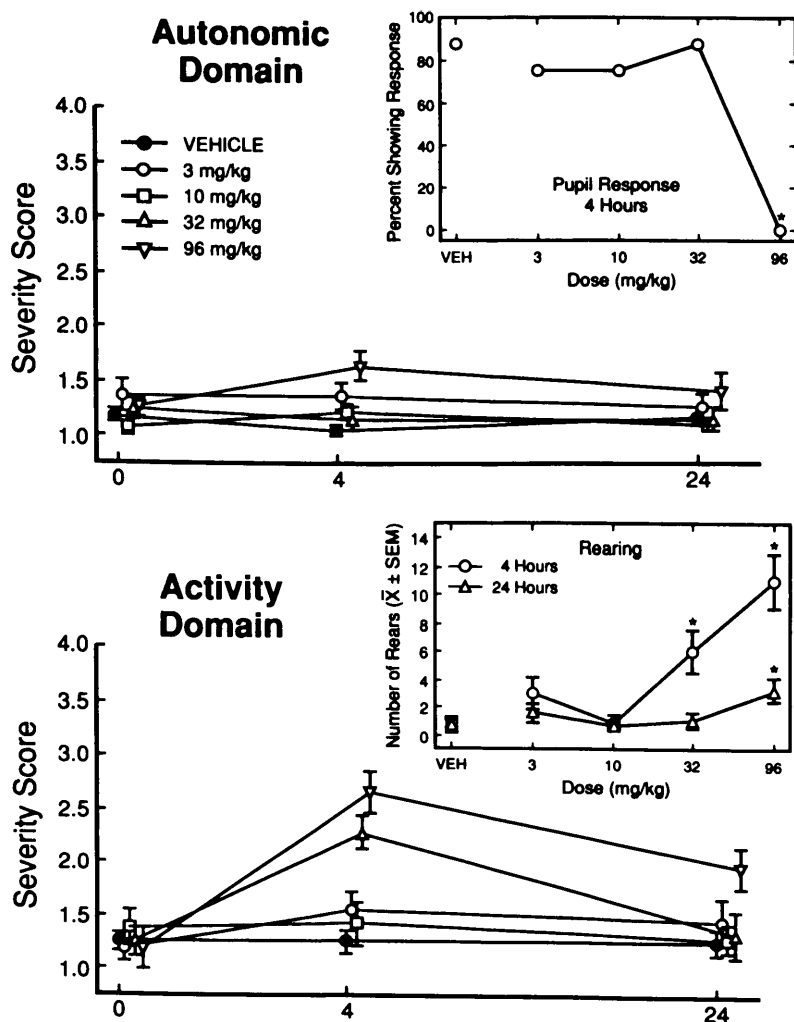


FIGURE 2. Dose response and time course of severity scores in each functional domain significantly affected by TDM (acute study). Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

that divergent changes (both increases and decreases) in the forelimb grip strength data of treated rats contributed to this finding.

Heptachlor

No lethality occurred with acute HEP. With repeated dosing, however, all rats receiving the two higher doses (23 and 69 mg/kg) died during dosing (the median number of doses to death were 3 and 6 for the highest and next-highest dose groups, respectively); in addition, one rat receiving the next

lowest dose (7 mg/kg) died shortly after the last dose. In the acute study the lowest dose (7 mg/kg) produced some significant changes, but with repeated dosing that dose level produced considerably more effects.

The magnitude of acute HEP effects on activity and excitability was greatest 4 h after dosing, and excitability changes were also observed at 24 h (Figure 3). Excitability changes included increased arousal and reactivity to removal from the home cage and to handling, with at least half of the rats in the two highest dose groups receiving the maximum score on these measures. Straub tail was also observed in these rats. In the two highest dose groups these effects were similar in magnitude at 4 and 24 h after dosing, whereas with the lower doses rats showed recovery. Changes in activity level occurred only in the highest dose group at 4 h, and included decreased motor activity and hunched posture in the home cage. The highest dose group also showed body weight loss (approximately 2%) and hypothermia (0.8°C decrease).

The profile of effects produced by repeated HEP administration consisted of altered activity, hyperexcitability, and autonomic effects (Figure 3). Significant effects were obtained at the 4-d test in the activity and excitability domains; no rats in the two highest dose groups survived until the 9-d test. Activity changes included increased rearing in the open field and decreased motor activity, and in the home cage HEP-treated rats displayed hunched posture. Excitability changes at 4 d included increased reactivity to removal and handling in the two highest dose groups, and an increase in arousal that

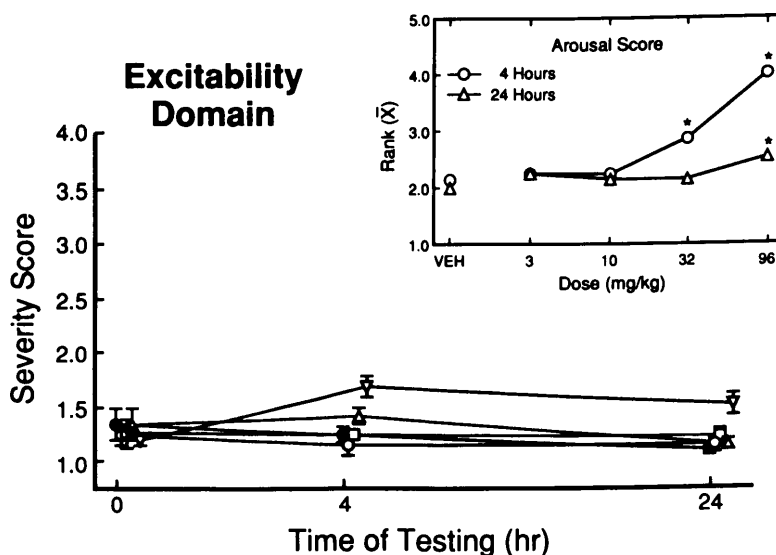


FIGURE 2. (Continued) Dose response and time course of severity scores in each functional domain significantly affected by TDM (acute study). Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

Acute

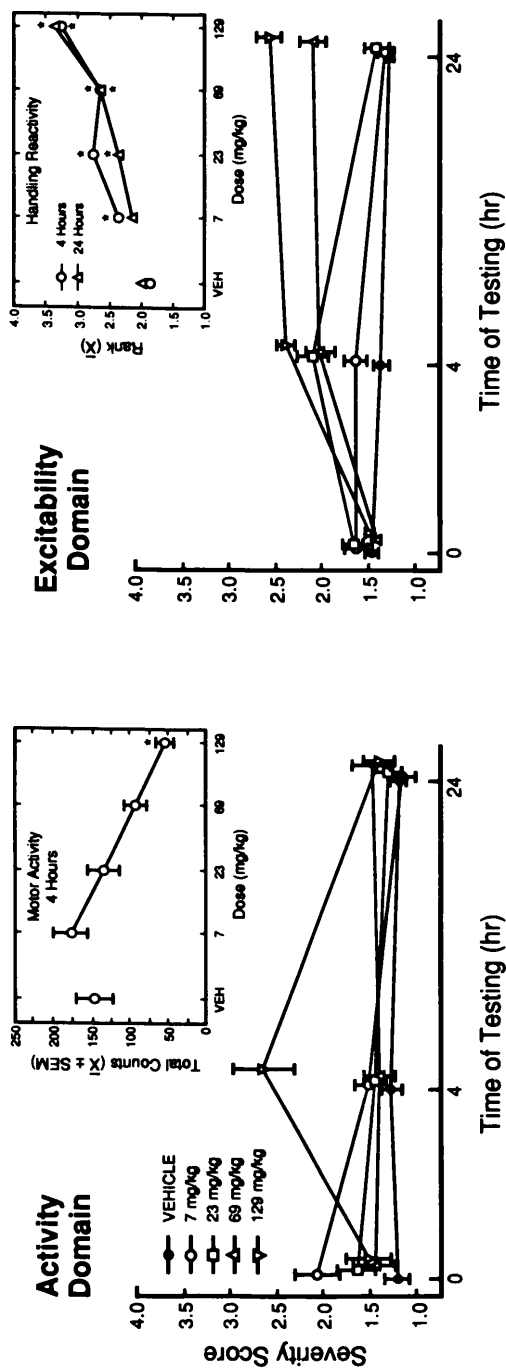


FIGURE 3. Dose response and time course of severity scores in each functional domain significantly affected by HEP. Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

Subacute

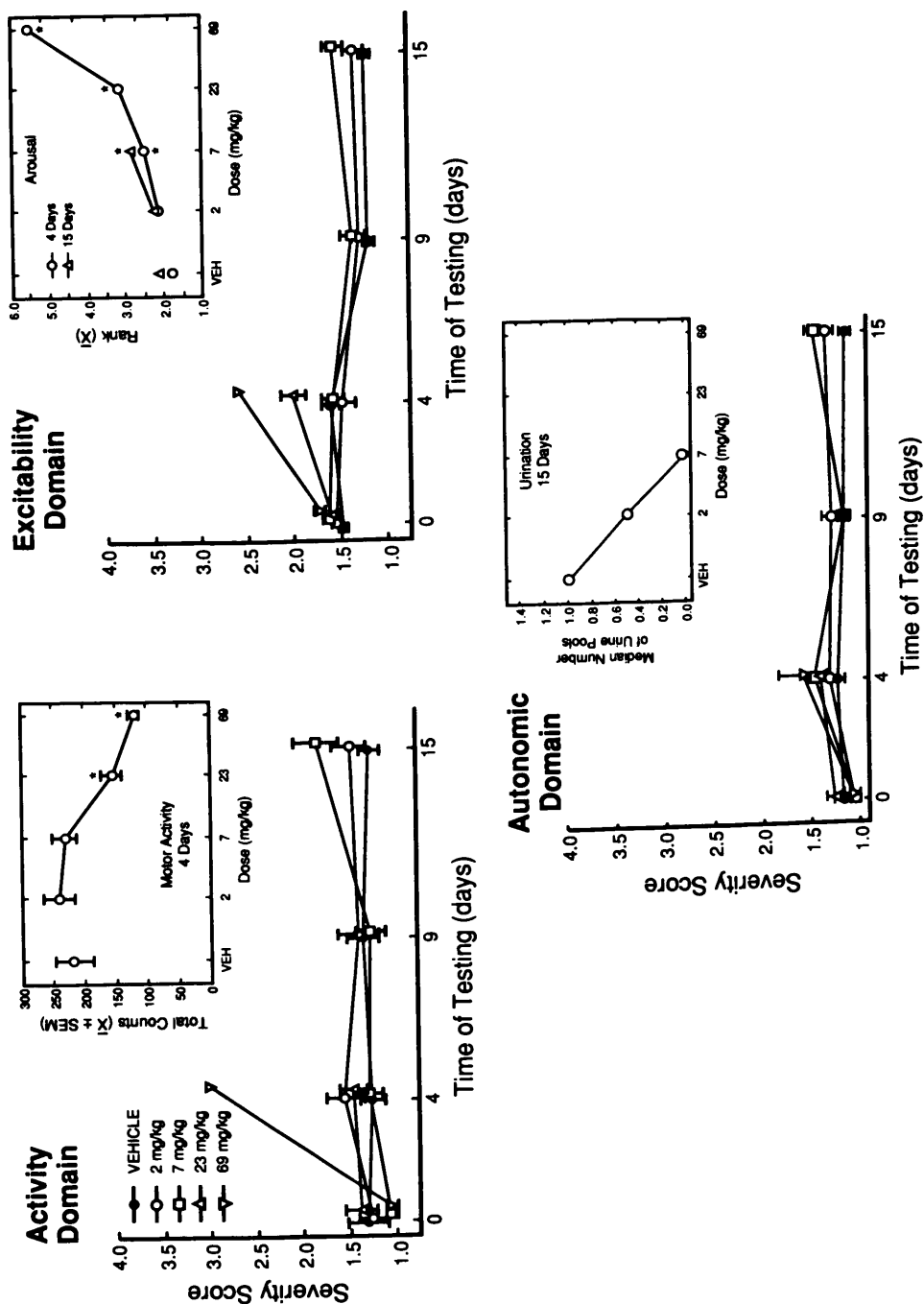


FIGURE 3. (Continued) Dose response and time course of severity scores in each functional domain significantly affected by HEP. Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

was significant in all but the lowest dose group. At 15 d, decreased urination in the highest surviving dose group contributed to the overall autonomic results; the excitability domain also approached significance ($p < .06$) in these rats. Hyperthermia (up to a 0.8°C increase) and weight loss (up to a 16% decrease) were also recorded before lethality occurred in the two highest dose groups.

Chlordane

No lethality occurred following acute CDN, but with repeated dosing all rats administered the two higher doses (52 and 156 mg/kg) died during dosing (the median numbers of doses to death were 4.5 and 10.5 for the highest and next highest dose groups, respectively).

Acute CDN produced significant effects in the excitability and neuromuscular domains (Figure 4). As with HEP, acute effects of CDN were most prominent 4 h after dosing, and some were still evident at 24 h. The excitability domain was significantly affected at both 4 and 24 h; rats in the two highest dose groups (156 and 291 mg/kg) displayed tenseness and increased reactivity to being handled. Clonic-tonic convulsions observed in some rats receiving the highest dose occurred at times other than the specified test times (2–3 h after dosing in two cases, and once during transport to the motor activity chambers). Neuromuscular function was significantly affected at 4 h; effects included tiptoe gait and increased forelimb grip strength in the highest dose group. Weight loss (up to 7%) was obtained in all but the lowest dose group, and piloerection was evident in the two highest dose groups.

Effects of CDN on activity and excitability were detected only in rats that eventually died with repeated exposure. CDN-induced changes in autonomic and sensorimotor domains were also significant (Figure 4). At 4 d, significant effects were obtained in the activity, excitability, and sensorimotor domains. These effects included decreased motor activity, increased removal and handling reactivity, and increased responsiveness to touch and auditory stimulation. In addition, weight loss was significant in the two highest dose groups. At 9 d, the excitability, sensorimotor, and autonomic domains were significantly affected; handling reactivity and arousal were still increased, and there was an increase in urination and lacrimation. The approach and tail-pinch responses were also increased at 9 d, while the touch response was decreased in the next lowest dose group (16 mg/kg). At the end of dosing, the autonomic domain was still significant, due largely to an increase in urination.

Phenol

In the acute study, two rats died within 4 h following the highest dose (224 mg/kg), and one rat died shortly after testing at 24 h after the next highest dose (120 mg/kg). In the subacute study, all rats in the highest dose group (120 mg/kg) died, but the time to death was protracted over the entire

Acute

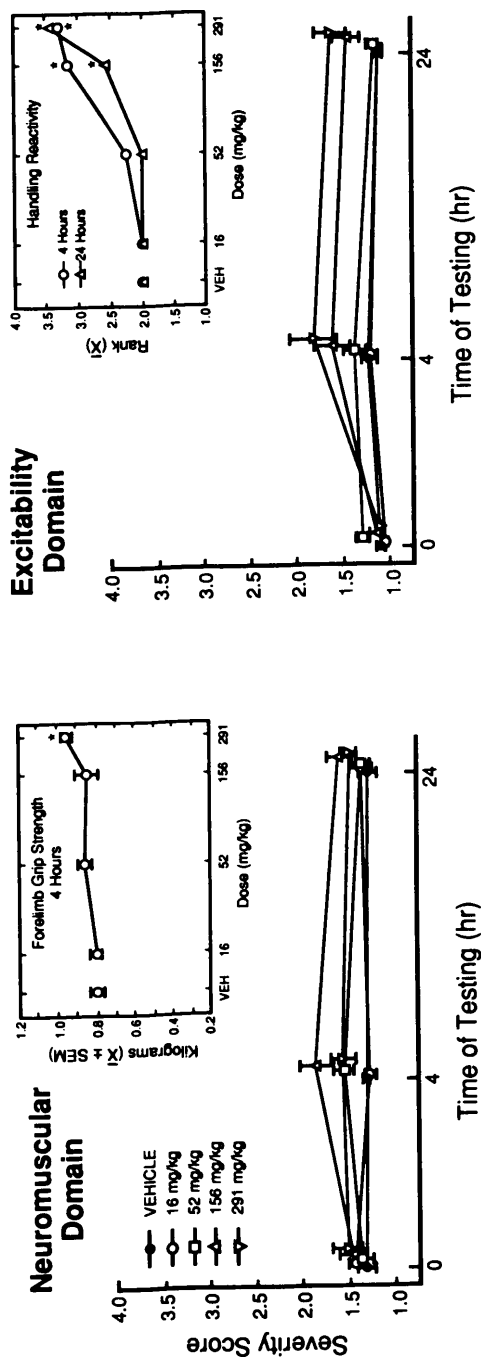


FIGURE 4. Dose response and time course of severity scores in each functional domain significantly affected by CDN. Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

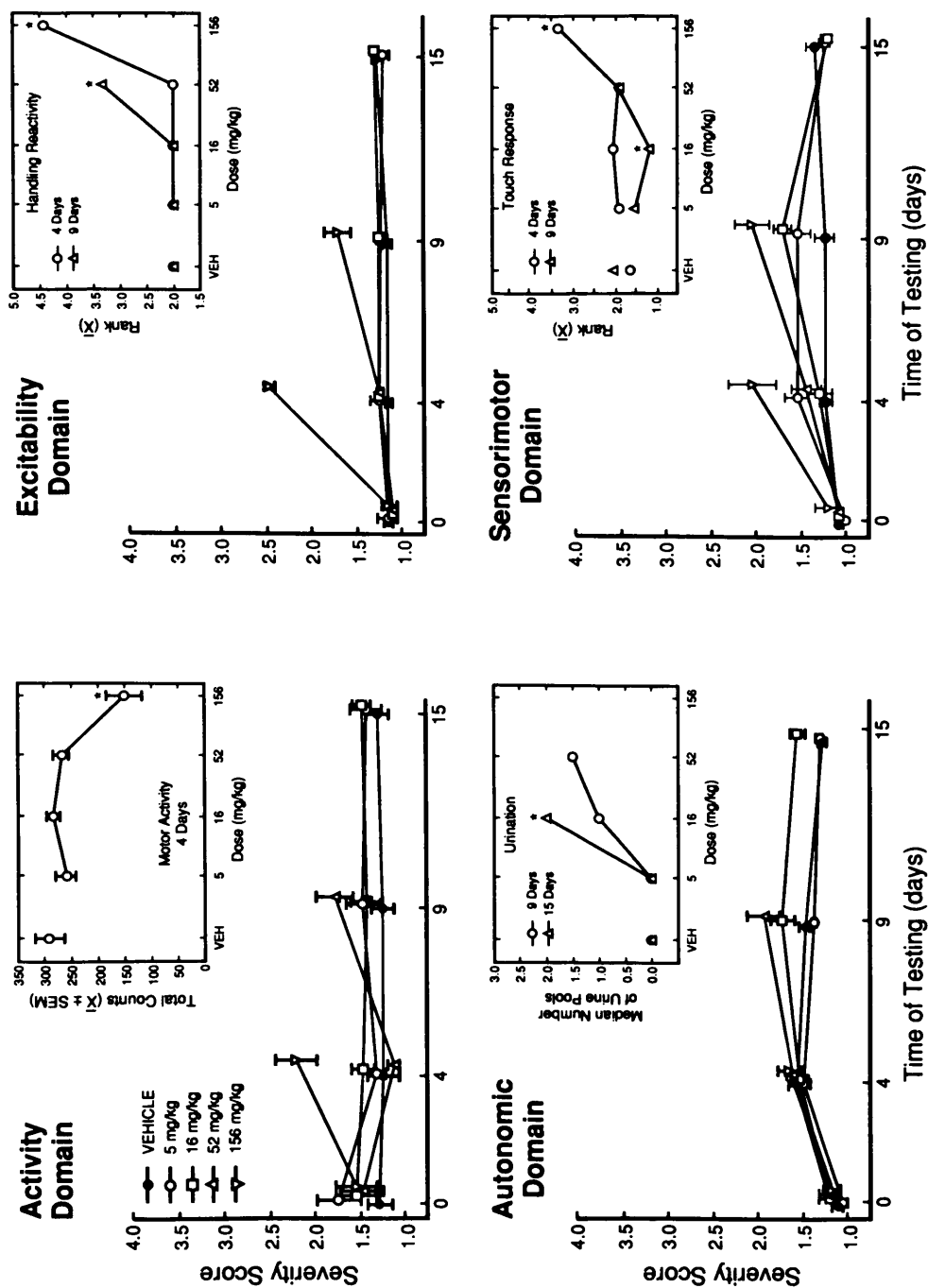


FIGURE 4. (Continued) Dose response and time course of severity scores in each functional domain significantly affected by CDN. Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

dosing regimen; the first death occurred after 1 dose, yet another rat received 11 doses before dying. This suggests that death was not due to an accumulation of phenol.

The most striking effect of phenol was tremors seen within 1–2 min after dosing. In the acute study, mild to severe whole-body tremors were seen in all rats receiving the two highest doses (120 and 224 mg/kg). No tremors were seen subsequently. Tremors were also seen in the subacute study immediately after the first administration of 120 mg/kg, but not after subsequent treatments. Convulsions were never seen in the lower dose groups, even with repeated dosing.

The activity and sensorimotor domains were significantly affected on the day of dosing with phenol (Figure 5). Effects in the two highest dose groups included decreased motor activity, crouched or flattened posture in the home cage, and decreased touch response. At 24 h, the neuromuscular domain was significantly affected by phenol; these effects included tiptoe gait and marginally increased landing foot splay in rats receiving the highest dose. Body weight loss (7–8% decrease) was also evident in the two highest dose groups at 24 h.

In the subacute study, autonomic function was significantly affected at 4 d, when inhibited pupil response was obtained in the highest dose group (Figure 5). The activity domain was significantly affected at 9 and 15 d. Slightly decreased motor activity levels were evident (but not significant), while increased rearing was obtained in the next highest dose group (40 mg/kg) at 15 d. A large (14%) decrease in body weight was observed in the highest dose group.

Diethylhexyl Phthalate

No lethality occurred in either DEHP study. Acute administration of the highest dose (5000 mg/kg) did, however, produce pronounced signs of general debilitation (ptosis, piloerection, slight lacrimation, and hypothermia) in two rats 24 h after dosing.

No functional domain was affected overall in either the acute or subacute studies. There were also no changes in body weight in either study.

Tetrachloroethylene

Four of eight rats treated with the highest acute dose of PER (5000 mg/kg) died within 24 h. No treatment-related deaths occurred in the subacute study. The highest dose showed pronounced effects in many of the FOB measures, whereas the next highest dose showed only about one-third of this activity.

Significant effects were obtained in the highest dose group on all functional domains 4 h after dosing (Figure 6). The next highest dose also significantly affected neuromuscular and sensorimotor function at 4 h. Autonomic changes in the highest dose group consisted of mydriasis, inhibited pupil response, lacrimation (also observed in the next highest dose group), and slight salivation. Decreased motor activity was obtained in the two highest

Acute

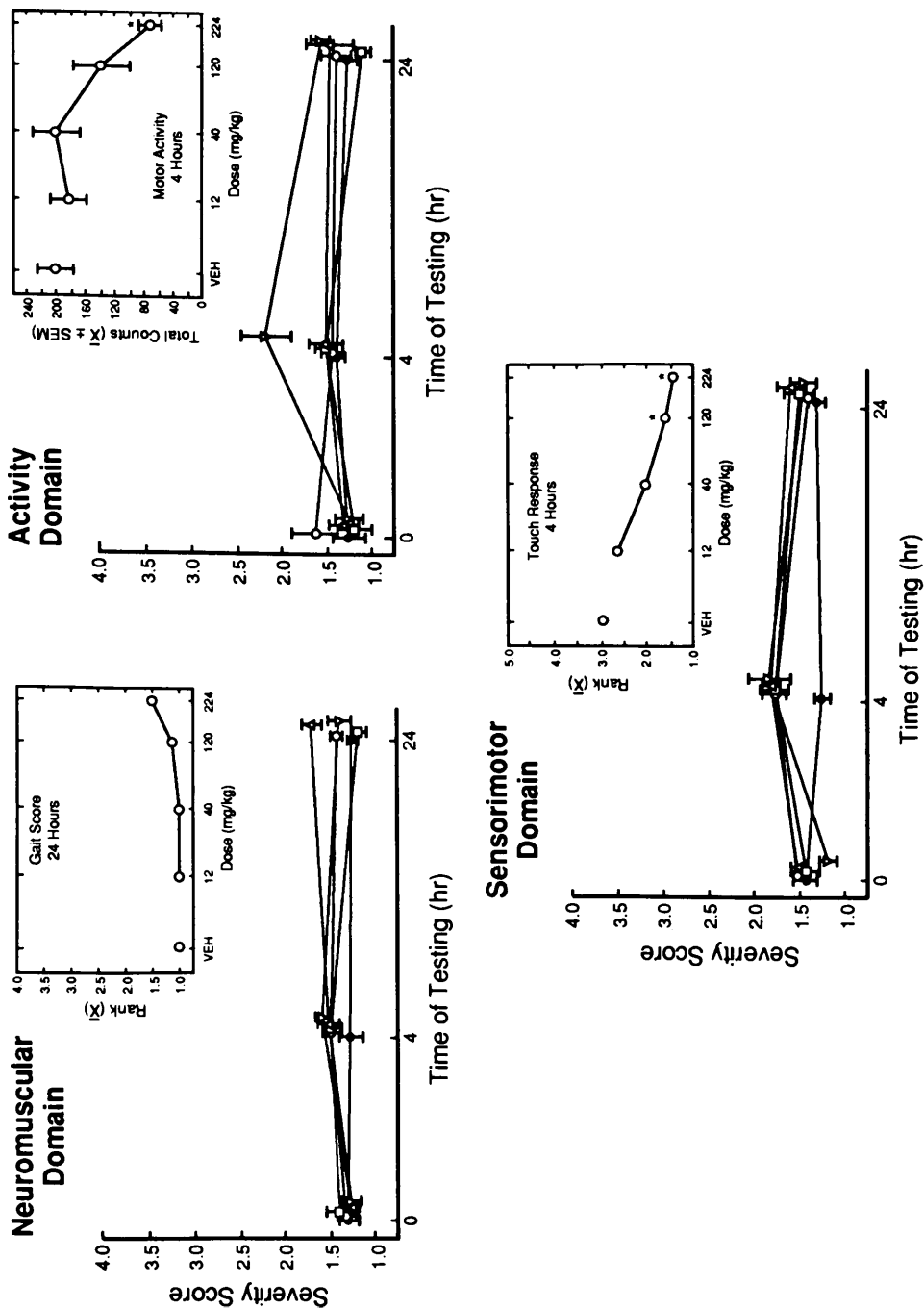


FIGURE 5. Dose response and time course of severity scores in each functional domain significantly affected by phenol. Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

Subacute

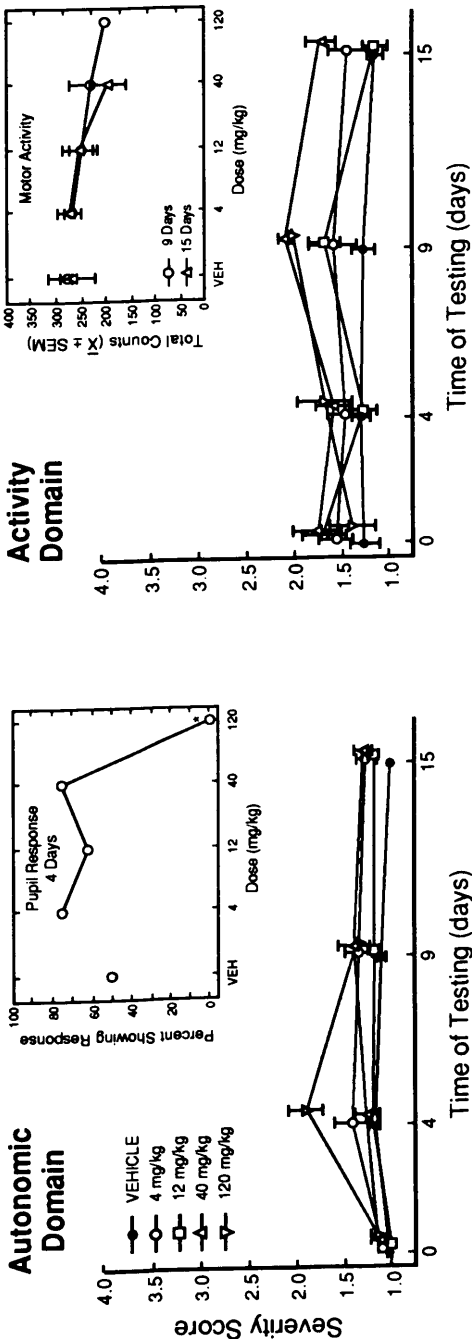


FIGURE 5. (Continued) Dose response and time course of severity scores in each functional domain significantly affected by phenol. Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

Acute

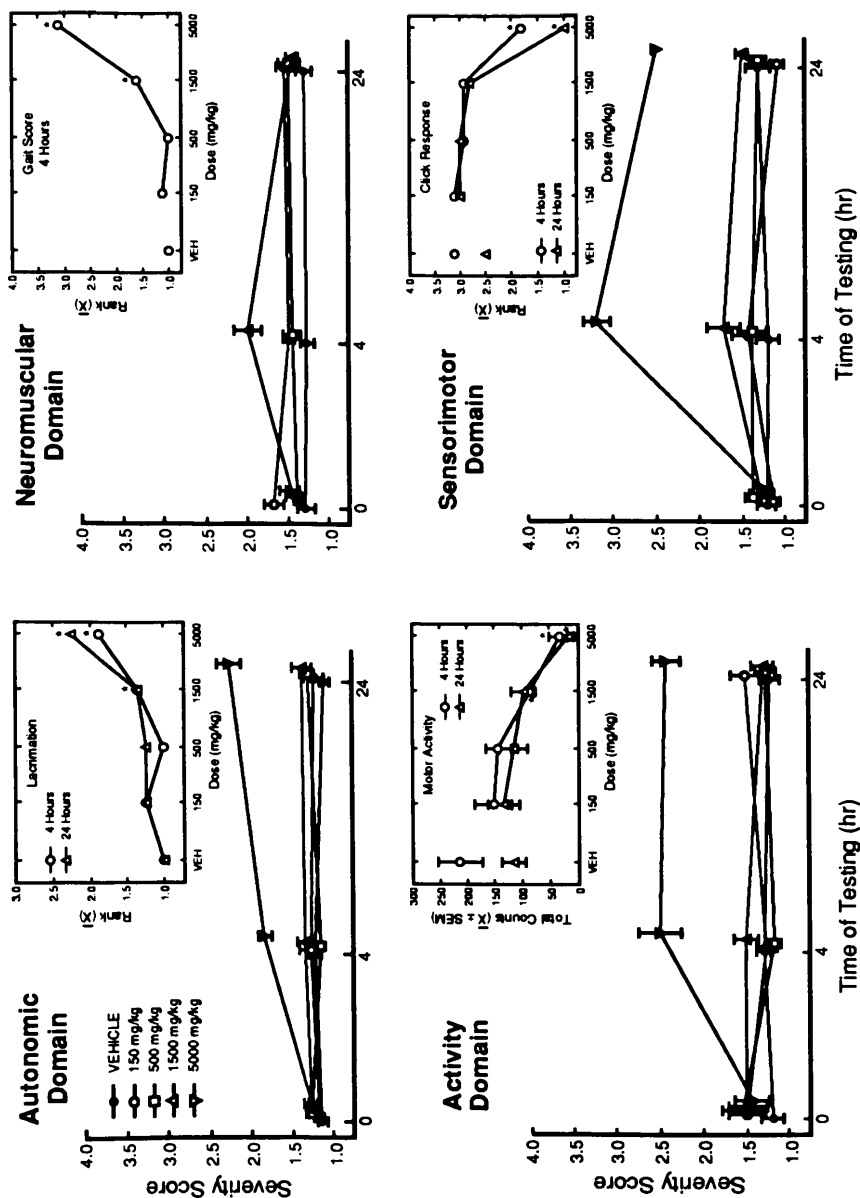


FIGURE 6. Dose response and time course of severity scores in each functional domain significantly affected by PER. Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

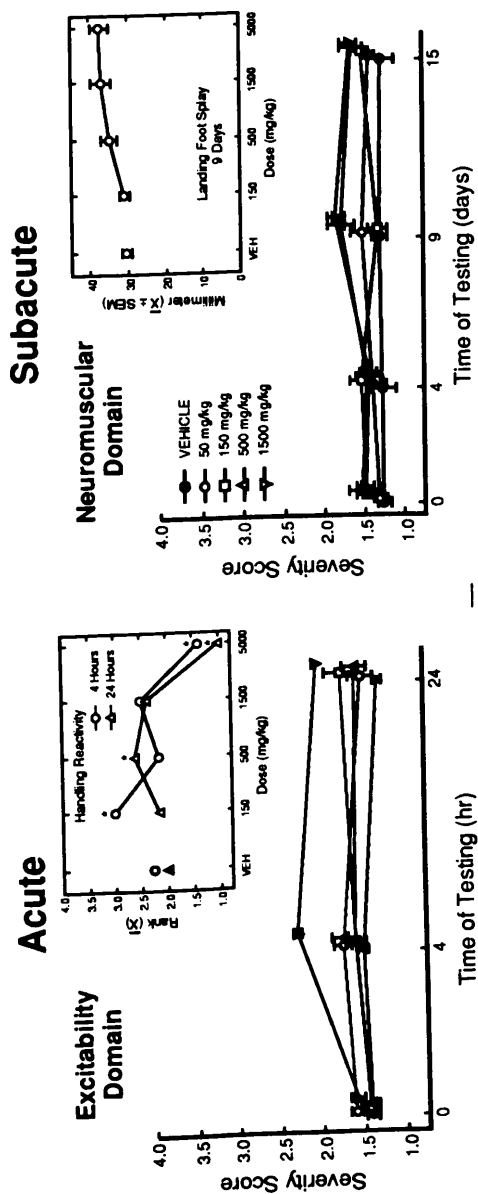


FIGURE 6. (Continued) Dose response and time course of severity scores in each functional domain significantly affected by PER. Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

dose groups, and flattened home-cage posture was seen in the highest dose group. Changes in excitability included decreased arousal and decreased removal and handling reactivity in the highest dose group. Increased arousal and handling reactivity were obtained in rats treated with lower doses, that is, an inverted U-shaped function. The two highest doses produced markedly abnormal gait and decreased righting ability; the highest dose also nearly doubled landing foot splay. In the highest dose group, grip strength data were not obtained at 4 and 24 h due to the rats' inability to grasp the device. Decreased responsiveness to all sensory stimuli were obtained in the highest dose group. These rats also displayed hypothermia (2.5°C decrease) at 4 h.

Most of the effects of PER at 24 h were obtained in the rats surviving the highest dose; for these rats, there was very little recovery from effects seen at 4 h. The autonomic, activity, excitability, and sensorimotor domains remained significantly affected. Autonomic effects of the highest dose included inhibited pupil response, slight salivation, ptosis, decreased defecation, and lacrimation (also in next highest dose group). The highest dose group showed decreased excitability (arousal and handling reactivity), while rats in the lower dose groups showed increases in removal and handling reactivity. Decreased responses to all sensory stimuli, except the approach of a blunt instrument, were also evident in the highest dose group at 24 h.

Only the neuromuscular domain was affected by repeated dosing with PER, but no individual measure was significantly affected (Figure 6). Body weights decreased in the highest dose group at 4 d, and although these rats did gain weight at 9 and 15 d they remained significantly lighter than control rats.

Trichloroethylene

No lethality occurred in either TCE study. Most TCE effects were obtained at 4 h in rats administered the highest dose (5000 mg/kg); some effects were also obtained at 24 h. There were relatively few effects attributable to repeated dosing with TCE.

Acute TCE significantly affected the activity, neuromuscular, and sensorimotor domains at 4 h (Figure 7). Significant effects in the highest dose group included decreased motor activity and abnormal home-cage posture. Neuromuscular changes in the high-dose rats included abnormal gait, increased landing foot splay, impaired righting, and decreased forelimb and hindlimb grip strength. These rats also displayed virtually no responsiveness to any stimuli except the approach of a blunt instrument. In particular, TCE decreased responsiveness to the click response at the next highest as well as highest dose. At 24 h, the activity domain was still significantly affected, due to decreased motor activity. Other signs of exposure included hypothermia (2°C decrease), weight loss (7%), and piloerection.

In the subacute study, the activity and neuromuscular domains were significantly affected at 15 d (Figure 7). Within the activity domain, only increased rearing in the next highest dose group was significant. Analysis of the neuromuscular measures did not reveal any significant treatment effects. Body weight loss was maximal at 4 d, and weight gain was evident at 9 and 15 d.

Acute

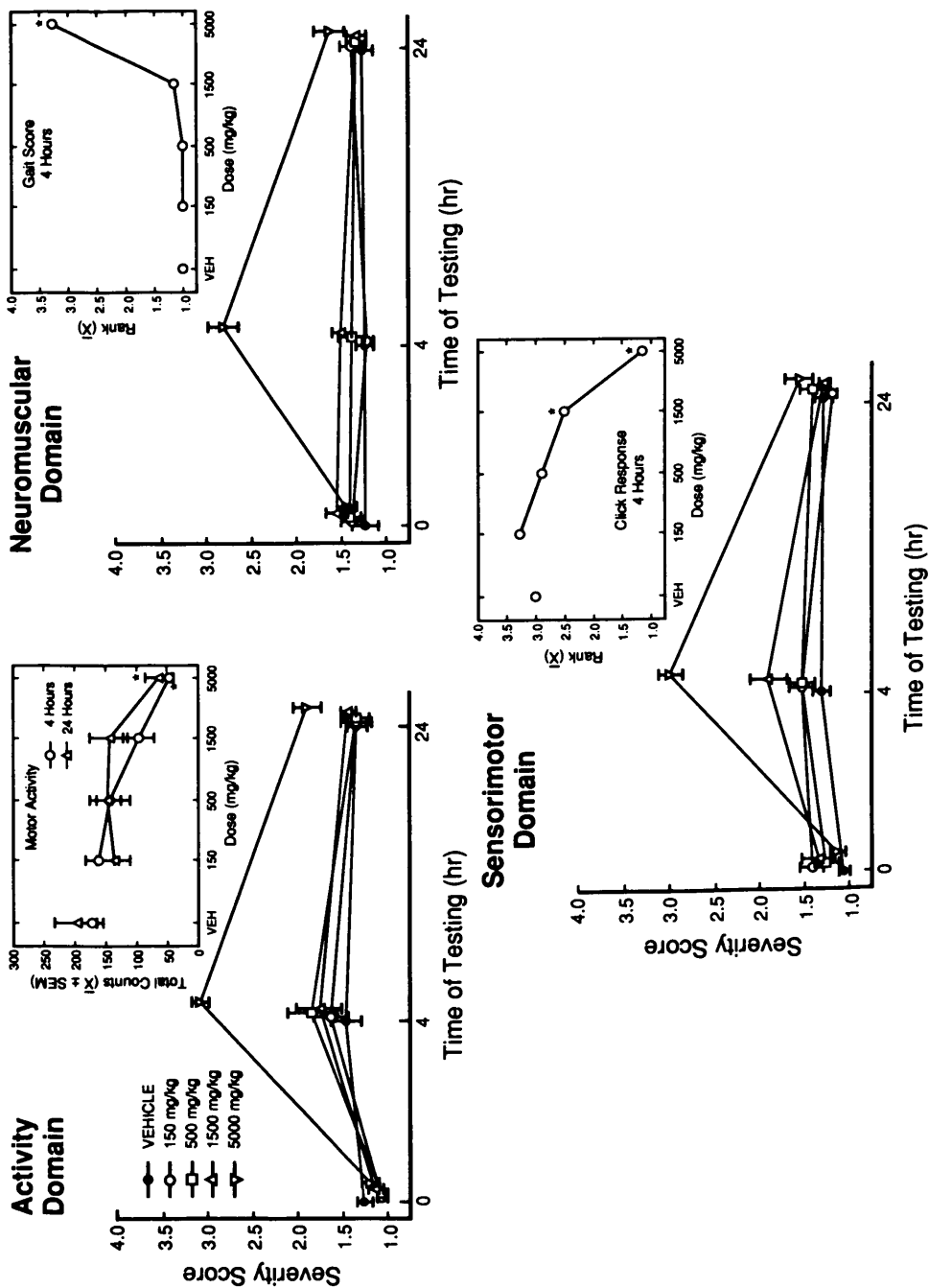


FIGURE 7. Dose response and time course of severity scores in each functional domain significantly affected by TCE. Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.



Subacute

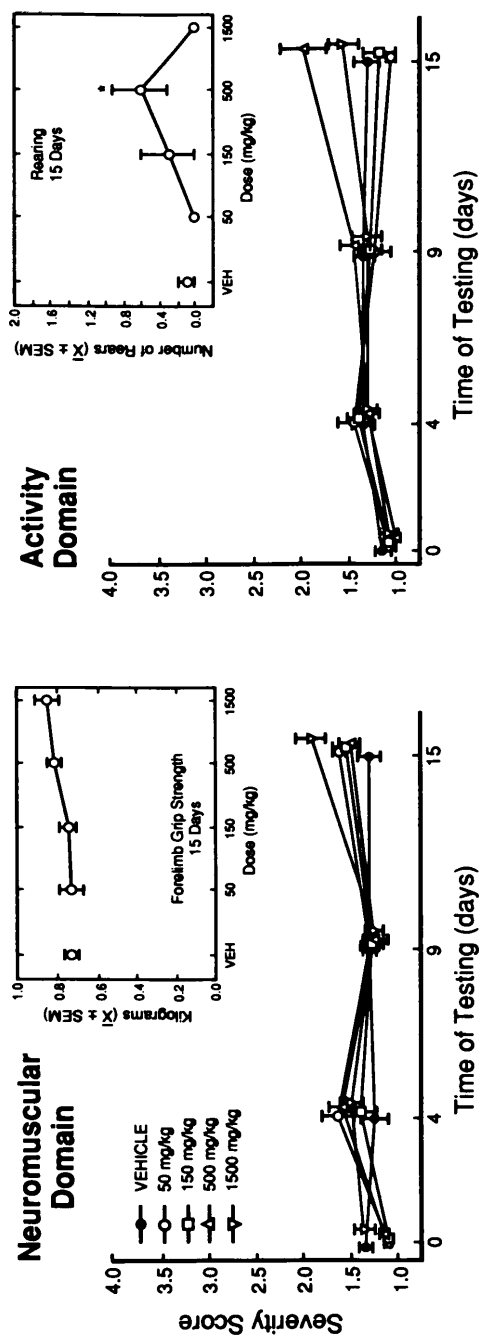


FIGURE 7. (Continued) Dose response and time course of severity scores in each functional domain significantly affected by TCE. Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

Carbon Tetrachloride

No lethality occurred in either CCl_4 study. In general, CCl_4 effects at 24 h were the same as or even greater than those at 4 h. In both studies the most sensitive endpoints were piloerection and weight loss.

The activity and neuromuscular domains were both significantly affected at 4 and 24 h after acute CCl_4 (Figure 8). Activity changes at 4 h consisted of

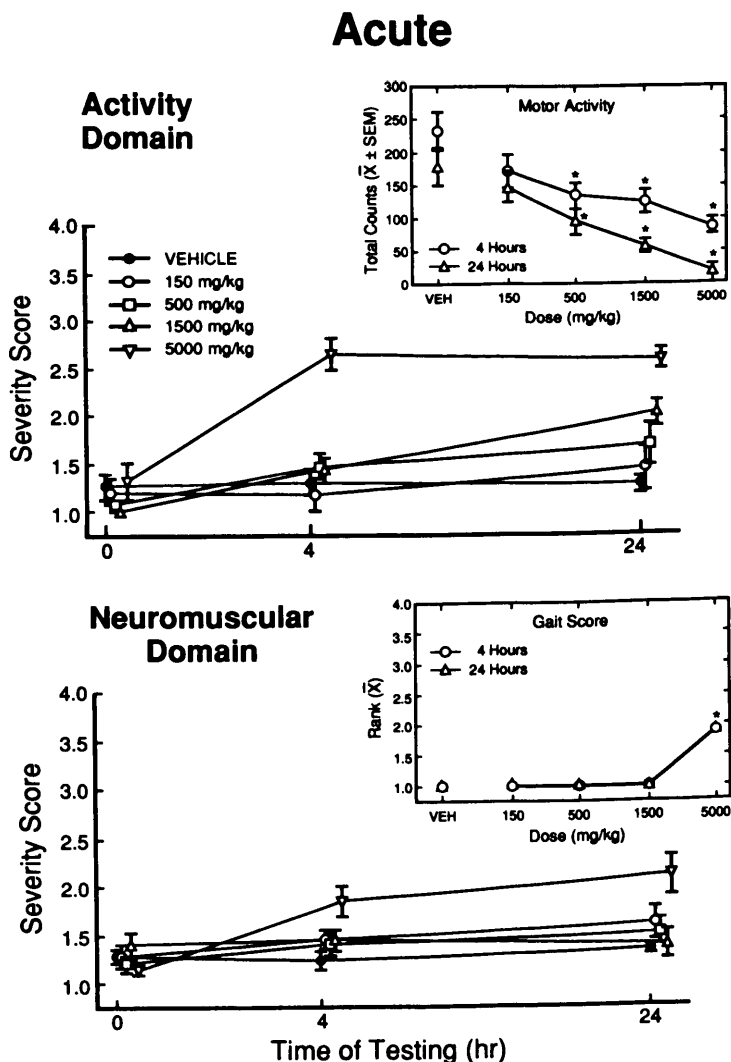


FIGURE 8. Dose response and time course of severity scores in each functional domain significantly affected by CCl_4 . Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

Subacute

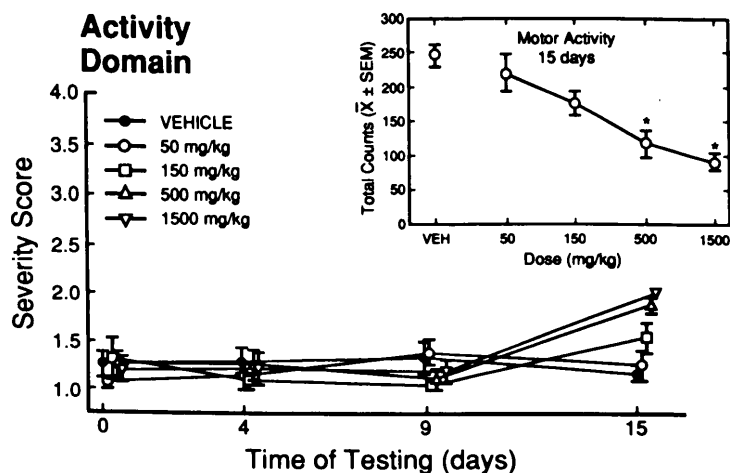


FIGURE 8. (Continued) Dose response and time course of severity scores in each functional domain significantly affected by CCl_4 . Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

curled posture in the home cage and increased open-field rearing in the highest dose group; decreased motor activity occurred in all except the lowest dose group. Rats in the highest dose group also displayed tiptoe gait. Rats in the two highest dose groups (1500 and 5000 mg/kg) appeared more affected at 24 h than at 4 h. Activity changes included curled posture in the home cage and decreased rearing in the two highest dose groups; motor activity was again decreased in all but the lowest dose group. Neuromuscular effects at 24 h included tiptoe gait with some ataxia, and marginally decreased hindlimb grip strength in the highest dose group only. Piloerection and body weight loss (up to 4%) were seen in all dose groups.

Repeated dosing with CCl_4 affected only the activity domain at 15 d, due to decreased motor activity. Repeated CCl_4 treatment also produced progressive loss of body weight in all dose groups, and piloerection in the highest dose group. Weight loss at the end of dosing ranged from 8% in the lowest dose group to 17% in the highest dose group.

Dichloromethane

No lethality occurred in the acute study of DCM. Seven deaths did, however, occur during repeated exposure to the highest dose (1012 mg/kg); time of death ranged from d 4 to 11. Acute DCM effects were obtained at 4 h primarily in the rats administered the highest dose. In the subacute study, most of the effects in the FOB were obtained in rats that subsequently died with continued dosing.

Neuromuscular and sensorimotor domains were affected at 4 h, but not

at 24 h after acute DCM (Figure 9). Rats in the highest dose group displayed ataxic and tip-toe gait, and decreased forelimb grip strength. Decreased reaction to the click stimulus was evident in the two highest dose groups.

In the subacute study, only three rats in the highest dose group (1012 mg/kg) were alive at 9 d, and only one survived for 15 d. At 4 d, the autonomic, neuromuscular, and sensorimotor domains were significantly affected by DCM (Figure 9). Neuromuscular effects in the highest dose group included altered gait, while autonomic effects included lacrimation, salivation, and inhibited pupil response. Responses to the click stimulus were also decreased in the two highest dose groups. At 9 and 15 d, the same neuromuscular and autonomic effects were evident in the survivors in the highest dose group. Hypothermia (2°C decrease) was also recorded in these rats at 4 and 9 d, along with a steady loss of body weight. Excitability was also affected at 9 d, but no effects on individual measures were significant.

DISCUSSION

Pesticides: Treatment Effects

Due to its cholinesterase-inhibitory properties, CAR typically produces signs of cholinergic stimulation (Cranmer, 1986). The effects obtained in this study were generally consistent with those reported in Moser et al. (1988). In the present study, the acute effects of CAR indicated autonomic activity, but the entire cholinergic syndrome (including diarrhea, polyuria, and salivation) was not observed. The most probable reason for this is that the 4-h time point was not optimal for this chemical. Our previous studies with CAR (Moser et al., 1988) indicated that at 90 min, effects of doses as low as 10 mg/kg (ip) were overt and partial recovery was evident by 3 h. In the present study, higher oral doses (78 and 145 mg/kg) were necessary for the effects to be observed at 4 h. The lack of activity in the subacute study of CAR was not unexpected, given the few residual effects observed at 24 h in the acute study (pupil response and lacrimation) and the relatively short half-life of carbaryl (Cranmer, 1986).

Of particular interest was the depression of the tail-pinch response, which was a sensitive indicator of CAR exposure and occurred at lower doses than those that produced generalized weakness. This decreased responsiveness specifically to the tail pinch, which was also observed by Moser et al. (1988), probably reflects antinociceptive properties, which have been characterized for cholinomimetics (Metys et al., 1969; Pedigo et al., 1975).

The effects of acute administration of TDM, in a different strain of rats (Long-Evans), have been previously reported (Moser & MacPhail, 1989). The profile of activity in these two studies is quite similar, and includes increased arousal, stereotypies, tremors, and hyperactivity. These effects closely resemble those produced by psychomotor stimulants such as *d*-amphetamine and

Acute

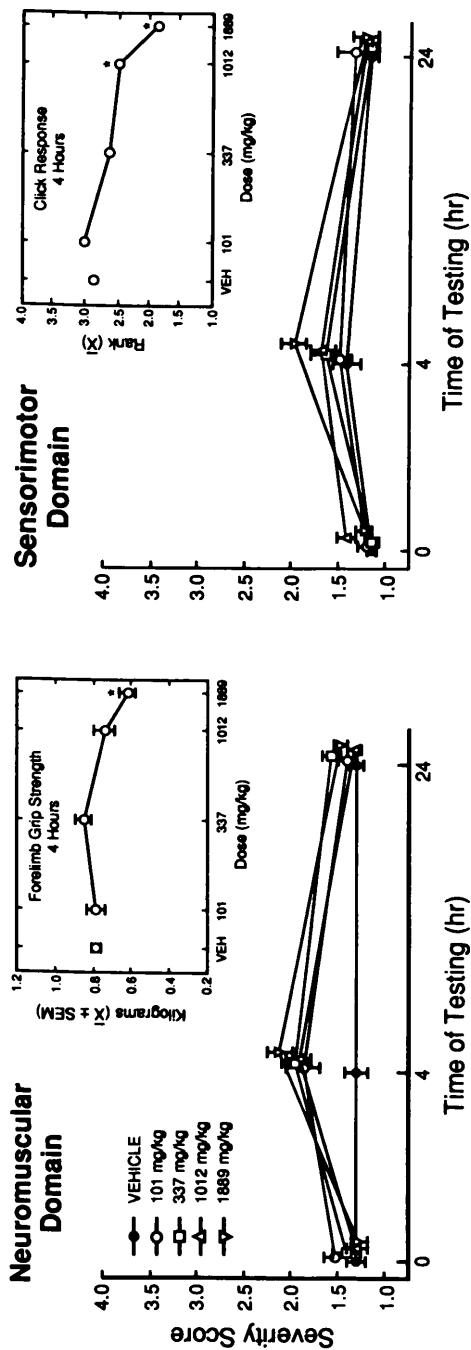


FIGURE 9. Dose response and time course of severity scores in each functional domain significantly affected by DCM. Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

Subacute

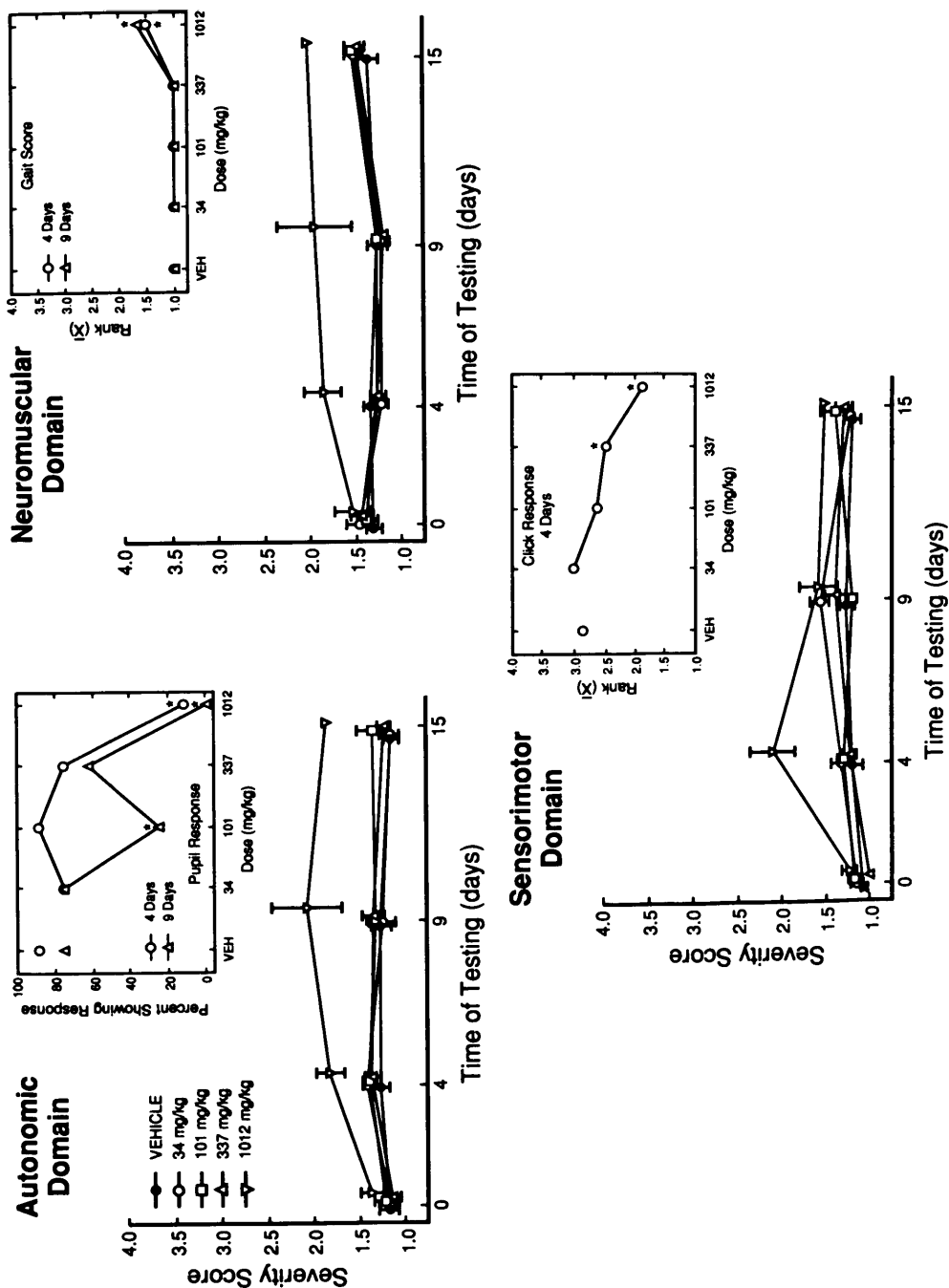


FIGURE 9. (Continued) Dose response and time course of severity scores in each functional domain significantly affected by DCM. Within each domain at each significant time, the dose response for the individual measure that showed the greatest effect is shown in the inset.

methylphenidate (Iversen & Iversen, 1981; Lyon & Robbins, 1975). The behavioral profile and some limited biochemical data implicate catecholamine neurotransmitters in these actions (Crofton et al., 1989; Moser & MacPhail, 1989; Walker et al., 1990).

A few differences were evident, however, in comparing the results of Moser and MacPhail (1989) to those of the present study. For example, hypothermia was produced by TDM in the earlier study, while hyperthermia was obtained in the present study. In addition, recovery from many of the acute effects of TDM was evident at 24 h in the present study, suggesting a fairly rapid elimination of the chemical. In the earlier study, however, the duration of TDM effects was longer (up to 48 h). Possible explanations for these differences include the strain of rat and/or the route of administration of TDM.

The chlorinated cyclodienes, CDN and HEP, produced hyperexcitability, especially increases in reactivity to removal and handling. Since control Fischer 344 rats often display low reactivity, these increases were especially notable. Treated rats also were more alert, tense, and showed straub tail. Acute effects were most prominent at 4 h, but many were still evident at 24 h after dosing. Tiptoe gait and increased grip strength were significant neuromuscular effects of CDN, and these findings have been obtained with other chemical classes that increase excitability (Moser et al., 1988; Moser, 1991a). Convulsions were observed with CDN, but not at the scheduled test times. Convulsions were not observed in the HEP study, but due to the limited times of observation it cannot be concluded that these compounds differed in this respect.

In contrast to increased excitability, motor activity was decreased by CDN and HEP in both the acute and subacute studies. The effects obtained in these studies correlate well with the reports in animals and humans of irritability, hypoactivity, and convulsions following acute exposure to cyclodienes [Agency for Toxic Substances and Disease Registry (ATSDR), 1989a, 1989b]. Furthermore, the decrease in motor activity along with increased excitability indicates that the pattern of effects produced by these cyclodienes is different from psychomotor stimulants. Effects on motor activity were thus a distinguishing feature in comparing the effects of these chlorinated cyclodienes with TDM.

Repeated dosing with CDN and HEP produced cumulative toxicity as evidenced by lethality in the two highest dose groups in both experiments. The increased reactivity to various stimuli appeared also to progress with repeated exposure; effects on handling, removal, arousal, and sensorimotor responses emerged and became evident at lower doses late in the dosing regimen. Changes in sensorimotor reactivity became marked during repeated dosing with CDN. This was not the case with HEP, but early deaths in the highest dose group prevented testing of all rats, and consequently it is not known which effects HEP would have produced in these rats before death occurred. CDN and presumably HEP are eliminated slowly from the body

(Barnett & Dorough, 1974; Ohno et al., 1986), which probably contributed to the cumulative toxicity and enhanced effectiveness with repeated dosing.

Pesticides: Neurobehavioral Profiles

Severity scoring analyses showed that acute administration of the pesticides produced different profiles of neurological effect. CAR and TDM produced clear effects on autonomic function. With CAR, evidence of lowered activity along with decreased responsiveness to stimulation indicated a generalized nervous system depression. On the other hand, TDM, HEP, and CDN all increased excitability. TDM and the cyclodienes could also be differentiated by the direction of the change in motor activity. Distinct profiles were also found with repeated administration of the pesticides. CAR and TDM had little or no effect, whereas cyclodienes showed cumulative effects. An interesting distinction between repeated exposure to the cyclodienes was the sensorimotor and autonomic effects of CDN, but not HEP. If these pesticides had been unknown compounds, these studies would have indicated acute neurobehavioral effects of concern. In addition, the cyclodienes would have appeared more persistent, possibly through accumulation in the body, produced progressive effects on the nervous system, and would thus pose more significant hazard with long-term use. Thus the overall assessments derived from these screening data agree with what is known about these pesticides.

Industrial Chemicals: Treatment Effects

The biological effects of DEHP include functional and morphological changes in liver and testes, birth defects, peroxisome proliferation, and carcinogenicity (Mitchell et al., 1985; Shaffer et al., 1945; reviewed in ATSDR, 1987). The hepatic and developmental effects of DEHP were verified in other portions of this study (Berman et al., 1995; Narotsky & Kavlock, 1995). No neurobehavioral measures were affected following either acute or repeated exposure to DEHP. Thus, in the present experiment DEHP produced toxic effects but no apparent neurotoxic potential.

The effects of phenol in human and laboratory animals have been reviewed by Liao and Oehme (1980). Acute signs of toxicity are similar in all species studied and are marked by neuromuscular hyperexcitability with muscle twitching progressing to severe convulsions. In addition, heart rate and blood pressure transiently increase and then decrease, and there may be salivation, dyspnea, and hypothermia. Tremors and convulsions were also reported in our studies, but other effects were not observed, probably because formal observations were not scheduled until 4 h after dosing. Two mechanisms have been proposed for these convulsions. One study reported the mode of action to be increased excitation of the motor portion of spinal anterior horn cells (Deichmann & Keplinger, 1963), while another study suggested increased acetylcholine release at the neuromuscular junction (Otsuka & Nonomura, 1963). The apparent tolerance to the convulsive effects that

developed with repeated dosing has not been reported elsewhere but was probably not due to metabolic changes, since phenol continued to produce toxicity (lethality) in that dose group.

Many reports have remarked on the rapidity with which phenol-induced convulsions occur (Angel & Rogers, 1972; Liao & Oehme, 1980, 1981). In our study, the convulsions began within 1–2 min of dosing. This indicates extremely fast uptake and distribution via the blood. Indeed, peak blood levels occur within minutes of administration (Deichmann, 1944), and phenol is almost completely excreted by 24 h (Capel et al., 1972). Death due to over-exposure usually occurs within a few minutes (Liao & Oehme, 1980). However, in our acute study, death occurred 4–24 h after dosing, long after the convulsions had ceased. Given the rapid elimination of phenol in the rat, it seems unlikely that death was due to a direct effect of exposure. Likewise, no accumulation of phenol would be expected with repeated dosing. Yet in the subacute study, all rats in the highest dose group died at various times during the dosing period. Degenerative changes in the liver and kidney may be partially responsible (Berman et al., 1995), but the cause of death cannot be determined from these data.

Industrial Chemicals: Neurobehavioral Profiles

Far less neurological data has been accumulated for phenol and DEHP than for the pesticides just discussed. The overall lack of effects of DEHP suggests that, in a screening context, further testing for neurotoxicity would not be warranted. In these studies, the effects of phenol were somewhat unexpected. The acute tremors, convulsions, and other effects are probably of neurological origin, and the subacute toxicity (lethality) would also be of concern in an overall assessment. Thus, the profile of phenol effects clearly indicated the potential for neurotoxicity, but the syndrome was not as clear as with the pesticides. Follow-up studies, including neuropathological, neurochemical, and/or functional assessments, may be warranted to better define the neurotoxic potential of phenol.

Solvents: Treatment Effects

The four organic solvents tested in these studies were not potent, requiring acute oral doses of 2000–5000 mg/kg to produce effects. However, this is typical of many oral-dosing studies of low-molecular-weight solvents, including alcohols. In the LD50 studies (Berman et al., 1995), no deaths occurred at 5 g/kg with PER, yet all rats dosed with the next lower dose (3.85 g/kg) succumbed. Thus it was difficult to obtain an accurate LD50 value for PER, and it is notable that this was the only chemical with significant deaths at the high dose in the acute study.

TCE and PER were the most similar of the solvents tested in these studies. TCE showed a consistent profile in that most depressant effects in the functional domains were obtained at the highest dose at 4 h and were not accompanied by lethality. Indeed, of these solvents, TCE exerts more general

anesthetic properties and was used widely for these purposes in the early 1900s (Hayes, 1982). TCE has been reported to adversely affect motor activity, swimming, and a variety of conditioned performances (Annau, 1981), and these behavioral tests have been shown to be sensitive to chemicals that produce generalized nervous system depression.

Repeated-exposure studies have produced some contradictory reports of subanesthetic effects in both animals and humans (Annau, 1981; Hayes, 1982). Auditory dysfunction, a midfrequency hearing loss, has been clearly established (Crofton & Zhao, 1993; Jaspers et al., 1993; Rebert et al., 1991); however, we could not detect auditory changes using our clicker (a mixed-frequency stimulus). Using some similar test measures, Kulig (1987) also showed that a much longer period of repeated exposure to TCE did not affect motor activity, forelimb and hindlimb grip strengths, or coordinated hindlimb movement. The present study indicates considerably fewer effects of repeated exposure compared to the acute study, but the activity and neuromuscular domains were still significant. The activity changes were slight, and no individual neuromuscular measure of that domain was affected (in agreement with Kulig, 1987). This could represent either a statistical false-positive finding (type I error), or slight effects on the measures that individually were not significant but that became pronounced when the small differences were summed.

PER also produced acute nervous system depression, but the highest dose also caused 50% lethality, whereas the next highest dose was not nearly as active. For example, the 1500 mg/kg dose did not decrease general and sensorimotor reactivity but did lower motor activity and produce weakness. PER also produced significant effects on autonomic function, in particular lacrimation. Although lacrimation has been reported previously for PER (Hayes, 1982), it has usually been attributed to the irritant properties of inhalation exposure. The present data suggest that PER may have systemic irritant or cholinergic properties as well. Although the mammalian toxicity of PER has been reviewed (Reichert, 1983; Seeber, 1989), few neurobehavioral studies have been conducted in laboratory animals. With repeated exposure, PER produced only slight neuromuscular changes.

An extensive literature exists on the toxic effects of CCl_4 , but most studies deal only with the hepatotoxic effects for which it is best known. In the present study as well, hepatic changes were produced by all doses, with fatty infiltration, necrosis, and large increases in a number of serum enzymes (Berman et al., 1995). Alterations in physiological measures (body weight, piloerection) at all doses most likely reflected this systemic toxicity. Acute neurobehavioral effects also were of a generalized depressant nature, with decreased activity and neuromuscular weakness; interestingly, sensorimotor function was not affected as with TCE and PER. CCl_4 , 24 h after a single dose of 1589 mg/kg, was previously studied using a battery of neurobehavioral tests (Gerber & O'Shaughnessy, 1986). Similar to our results for a dose of 1500 mg/kg, those authors reported decreases in motor activity (time of

horizontal movement) and body weight loss, but no effect on forelimb or hindlimb grip strength or landing foot splay.

Neurotoxic effects of DCM have been reviewed (ATSDR, 1989c; Winneke, 1981), and in the rat these include decreased wheel-running activity, reduced REM sleep, and slowed conduction velocity in sciatic nerve. Although these effects could be due to acute nervous system depression, a confounding factor with DCM is the metabolic production of carbon monoxide (CO; ATSDR, 1989c). The relative importance of depressant activity and CO-induced hypoxia is not well understood as yet. In the present study, DCM was the least like the other solvents, which may be related to the unique involvement of carboxyhemoglobin. However, these tests were also conducted at lower dose levels than with the other solvents. For instance, the highest acute dose of DCM (1889 mg/kg) significantly affected about the same number of individual measures as did the 1500 mg/kg dose of the other solvents. The nature of these changes was somewhat different and, unlike the other solvents, acute administration of DCM did not alter the activity domain. Instead, the clearest effects were on the neuromuscular (altered gait and decreased forelimb grip strength) and sensorimotor (decreased click response) domains. In humans, DCM produces behavioral changes characteristic of depression and changes in auditory function (Winneke, 1974); auditory effects have been reported in rats as well (Rebert et al., 1989). The marked behavioral changes and lethality obtained with repeated dosing indicate that DCM may have more toxic potential with repeated exposure than do PER and TCE. As with phenol, a protracted lethality curve was evident, indicating that death was not due to some critical body burden, or threshold, of the chemical. The emergence of neurobehavioral effects with repeated dosing of DCM was unexpected, and warrants further study.

Solvents: Neurobehavioral Profiles

The acute effects of TCE, PER, and DCM were most evident 4 h after dosing, whereas the effects of CCl₄ were of equal or greater magnitude 24 h after dosing than at 4 h. For PER and TCE, and to a lesser extent CCl₄, the overall profile of acute effects was one of generalized nervous system depression, with a cluster of effects indicating neuromuscular weakness, lowered activity, and decreased responsiveness to stimulation. In general these effects were obtained at high dose levels and are typical of high-dose effects seen with many other solvents (Klaassen, 1985). In humans, the primary signs of acute exposure to organic solvents include drowsiness, dizziness, incoordination, and altered vigilance and motor performance, progressing to stupor (Hayes, 1982; Klaassen, 1985). Thus, most of the effects reported herein are consistent with the signs and symptoms reported in humans. Interestingly, lower doses of PER, TCE, and especially CCl₄ produced increases on some measures of excitability. These biphasic effects are probably due to depression of inhibitory influences, typical of central depressants.

While the overall profiles of these solvents appeared similar, closer inspection of the data did show differences in their acute effects. In a screening context, however, these acute effects would flag these chemicals as being potentially dangerous in high-level exposure scenarios. With PER, TCE, and DCM, the reversibility of these effects at 24 h indicates less concern for residual effects of acute exposure. CCl_4 , on the other hand, displayed marked effects at 24 h and thus may be singled out as a higher priority concern.

On the other hand, following repeated exposure there were relatively few residual effects of TCE, PER, or CCl_4 , but the effects of DCM were greater. The development of tolerance was implicated with these first three solvents. For example, the excitability effects of PER (1500 mg/kg) at 24 h in the acute study were not evident at that dose level 23 h after the 14th dose. The attenuation of weight loss during repeated dosing of TCE and PER may also indicate some sort of tolerance or compensation. The data for DCM, however, imply that accumulation or sensitization may have occurred, since effects on the click response, for example, became more pronounced at lower doses in the subacute than in the acute study. Overall, the relative lack of effects of these solvents on the various functional domains with repeated dosing would indicate less concern for long-term exposures. The marked physiological changes with CCl_4 , however, would single out that chemical as having higher potential for toxicity, though probably not neurotoxicity.

General Discussion

Of these 10 chemicals, the pesticides were the most active in the neurobehavioral tests. In the acute studies, pesticide effects were detected at doses in the lower end of the relative dose range (usually <10% of the LD50 value) and were most prominent in altering nervous system excitability and activity levels.

The chemicals varied substantially in their acute and subacute profiles of effects. On the one hand, TDM and CAR showed short durations of action and few effects with repeated exposures, and on the other hand, the cyclodienes (CDN and HEP) had more prolonged effects and increasing toxicity with repeated exposures. Repeated dosing with three of the solvents (PER, TCE, and CCl_4) showed few neurobehavioral effects, whereas DCM showed more effects as well as lethality with repeated dosing. Finally, the industrial chemicals (DEHP and phenol) showed similar profiles of effects in both studies. Thus, in this sample of 10 chemicals, the acute studies of 80% of the chemicals were insufficient to predict results of repeated dosing. This finding emphasizes the necessity of both types of studies in toxicological assessments.

The predetermined test times and dosing regimens specified in these studies were not optimal for many of the chemicals, and it was difficult to predict the lethality obtained with repeated exposure. For the cyclodienes this was not unexpected, since these chemicals are known to accumulate

with continued exposure. However, neither phenol nor DCM showed appreciable effects at 24 h in the acute study, and kinetic studies suggest rapid elimination, yet both produced lethality with repeated dosing. These results indicate that, for almost all chemicals, pilot studies are irreplaceable for determining the time of peak effect and cumulative toxicity.

Finally, the present study, along with the companion reports, provides important information regarding these chemicals by establishing dose-response and time-course characteristics, using acute and subacute exposures, of three major forms of toxicity. This neurobehavioral test battery was capable of detecting neurological effects of these compounds. All the chemicals known to act on the nervous system scored as positive; that is, there were no false negatives. Applying the severity scoring scheme provided profiles that also correlate well with what is known about these chemicals. These results support the use of these or similar tests in the routine screening of a variety of compounds for neurotoxicity.

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